1 The what and why of the data mesh

This chapter covers

- Defining data mesh
- Introducing the key concepts of the data mesh paradigm
- Understanding why the data mesh is a socio-technical paradigm shift
- Seeing the advantages of the data mesh
- Identifying possible data mesh implementation challenges

The *data mesh* is a decentralization paradigm. It decentralizes the ownership of data, its transformation into information, as well as its serving. It aims to increase the value extraction from data by removing bottlenecks in the data value stream by these means.

The data mesh paradigm is disrupting the data space. Large and small companies are racing to showcase *their data mesh–like journey* all over the internet. It's becoming the new *thing* to try out for any company that wants to extract more value from its data. This book describes the data mesh paradigm as a socio-technical architecture, with an emphasis on the *socio*. The main focus is on people, processes, and organizations, not technology. Data meshes can, but don't have to, be implemented using the same technologies most current data systems run on.

But because the data mesh is a topic of ongoing debate, with only slowly emerging best practices and standards, we found the need for an in-depth book that covers both the key principles that make data meshes work and examples and variations needed to adapt this to any company. This book is designed to do just that: help you begin your own data mesh journey.

We will provide you with conceptual tools, including, but not limited to, the key data mesh principles and boots-on-the-ground insights and examples. This book is not meant to be a comprehensive theoretical study but offers you the practical guidance necessary to get set up.